Multiple Regression and the Hat Matrix — Restaurant Tips

From one predictor to two, and the matrix behind the fit

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https://github.com/HelloJJJie/Assignment-Forecast-and-Analysis-of-Regression.git

A dataset of $n = 244$ restaurant bills — what do the bill *and* the party size tell us about the tip?

1. The model in matrix form, and the fit

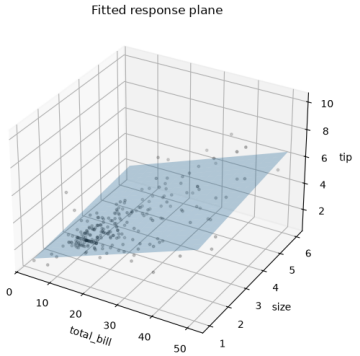
Data. $Y =$ tip, $X_1 =$ bill (USD), $X_2 =$ party size (1–6); $p = 3$.

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon \;\Rightarrow\; \mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

$\hat{Y} = 0.669 + 0.0927X_1 + 0.1926X_2$, $MSE = 1.027$ (241 df).

	b	$s\{b\}$	t^*	P
intercept	0.6689	0.1936	3.46	6.5×10^{-4}
total bill	0.0927	0.0091	10.17	1.9×10^{-20}
size	0.1926	0.0853	2.26	0.025

- **Partial effect:** b_1 is 11.7% below the simple slope 0.1050 ($r_{X_1X_2} = 0.598$).
- **Overall F:** $105.95 > F(0.95; 2, 241) = 3.03$, reject $\beta_1 = \beta_2 = 0$; $R^2 = 0.468$, $R^2_a = 0.463$ (bill alone: 0.457).



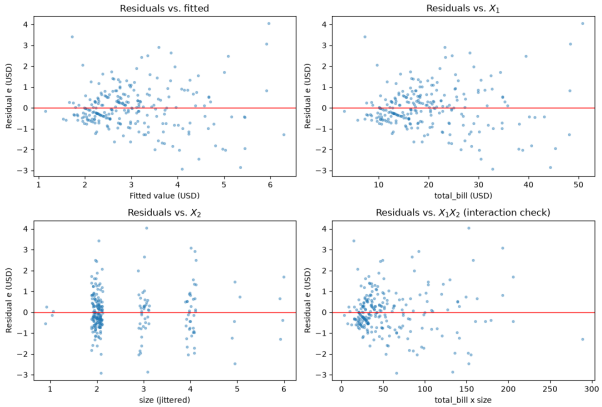
The response surface is a plane; each coefficient holds the other predictor fixed.

2. Prediction and diagnostics

Estimation at $\mathbf{X}_h = (1, 20, 3)$ ($20 bill, party of 3); $s^2\{\hat{Y}_h\} = MSE\, \mathbf{X}_h'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_h$:

fitted tip $\hat{Y}_h$	3.10
95% CI, mean tip	[2.955, 3.247]
95% PI, one new tip	[1.099, 5.103]

- **Constant variance rejected:** Breusch–Pagan $\chi^2 = 92.2$, $P = 9.4 \times 10^{-21}$; residual sd 0.68/0.76/1.42 by thirds of fitted value.
- **Normality rejected:** plot $r = 0.982$ vs. critical 0.994.
- **No interaction signal:** $\mathrm{corr}(e, X_1X_2) = +0.007$.



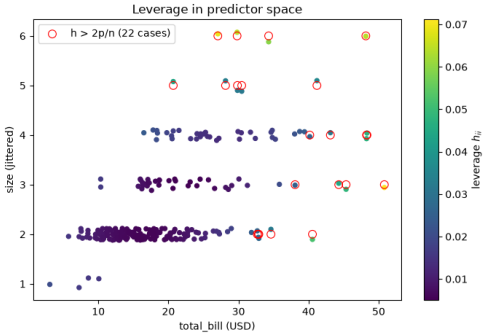
The funnel is the same one seen with a single predictor: the prediction interval has constant width, the scatter does not.

3. The hat matrix and leverage

$$\hat{\mathbf{Y}} = \underbrace{\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'}_{\mathbf{H}}\,\mathbf{Y}, \qquad \mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$$

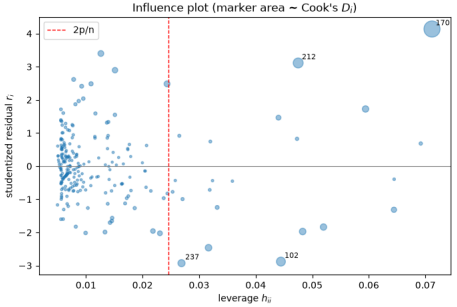
$\mathbf{H}$ is 244×244: a *projection* of $\mathbf{Y}$ onto the columns of $\mathbf{X}$. Checked numerically (max deviation $< 10^{-13}$):

- **Symmetric and idempotent:** $\mathbf{H}' = \mathbf{H}$, $\mathbf{H}^2 = \mathbf{H}$; $\mathbf{HX} = \mathbf{X}$, $\mathbf{X}'\mathbf{e} = \mathbf{0}$.
- $\operatorname{tr}\mathbf{H} = 3 = p$, so the average leverage is $p/n = 0.0123$; $SSTO = SSR + SSE$ $(465.2 = 217.7 + 247.6)$.
- **Leverage** h_{ii}: 22 of 244 cases exceed $2p/n = 0.0246$. The largest: bill \$50.81 $(h = 0.071)$; parties of 6 also rank high.
- $s^2\{\hat{Y}_i\} = MSE \cdot h_{ii}$: the mean-response interval is widest where leverage is highest.



Leverage is distance in predictor space, not in Y. A point can lie inside each variable's range and still outside the cloud: (50, 1) has $h = 0.138$, about $2\times$ the largest observed.

4. Residual variance and influence



Drop rows 170, 212, 102	b_0	b_1	b_2
change	+15.6%	−12.0%	+19.2%

- $\mathrm{Var}\{e_i\} = \sigma^2(1 - h_{ii})$: high-leverage points have *smaller* residual variance. The studentized residual $r_i = e_i/\sqrt{MSE(1 - h_{ii})}$ treats cases fairly — here it barely moves ($r_i/e_i^* \in [1.003, 1.038]$) because all h_{ii} are small.
- **Leverage is only potential influence.** Row 170 (bill \$50.81, tip \$10) combines the highest leverage with the largest residual: $r = 4.14$, Cook's $D = 0.437$ (next: 0.161).
- Dropping the three cases with the largest D shifts the coefficients by 12–19%.

Takeaway. The two-predictor model passes the overall F test but explains only 47% of the variance, fails constant variance and normality, and three cases move the coefficients by 12–19%.